



Distributed Information Processing

5th Lecture

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Outline

- Communication
 - Layered Protocols
 - Middleware Protocols
 - Types of Communication
- Communication Methods
- Q&A



Message Passing Requirements

■ Agreements Needed at a Variety of Levels

- Meaning of the Bits Being Sent
 - Character coding: e.g., EBCDIC and ASCII
- Number of Volts for a 1-bit
- Indication of the Last Bit of the Message
- Detection of Damaged or Lost Messages
- Lengths of Numbers, Strings, and Others
- Representations

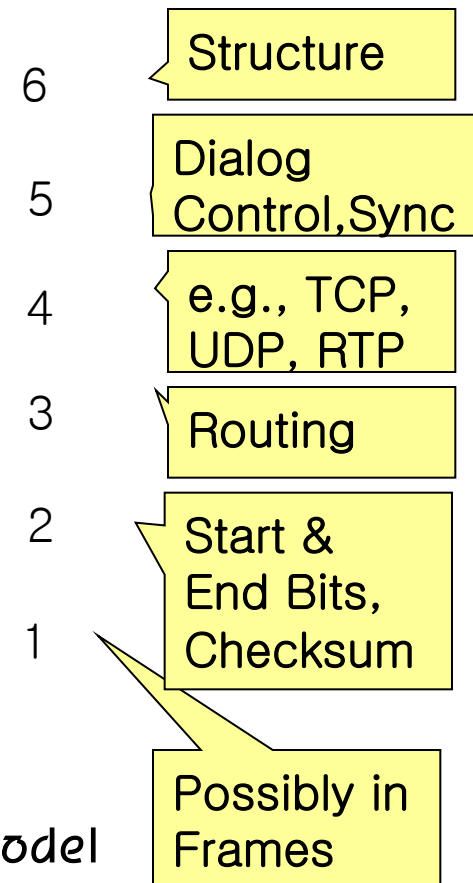
Agreements from the low-level details of bit transmission to the high-level details of how information is to be expressed

Layered Protocols

- ISO OSI (Open Systems Interconnection) Reference Model
 - Designed to Allow Open Systems to Communicate
 - Open system is prepared to communicate with any other by using standard rules that govern the format, contents, and meaning of messages
 - Protocols: such rules formalized
 - Connection oriented
 - Connectionless
 - Useful for Understanding Computer Networks
- Protocol Suite (or Stack)
 - Collection of Protocols Used in a System

Illustration: Layered Protocols (1)

Figure 4-1 on Page 118



Layers, interfaces, and protocols in the OSI model



Illustration: Layered Protocols (2)

Figure 4-2 on Page 118

A typical message as it appears on the network



Middleware Protocols

Figure 4–3 on Page 123

An adapted reference model for networked communication

Types of Communication

Figure 4-4 on Page 124

*Viewing middleware as an intermediate (distributed) service
in application-level communication*

Persistent vs. Transient Communication
Asynchronous vs. Synchronous Communication



Communication Methods

- RPC (Remote Procedure Call)
 - Communication by Calling Remote Procedures
 - Definition of service interface
 - Lack of ability to create new object instances
 - Lack of support for remote object references
- RMI (Remote Method Invocation)
 - Communication by Calling Methods of a Remote Object
 - Implementation of a remote interface
 - Creation of new object instances
 - Support for remote object references



Communication Methods (Cont'd)

■ Socket

□ Communication of Messages and Data between Processes

- Use of a raw communication channel
- Definition of a low-level message protocol
- Definition of data transmission format

■ Distributed Event-Based Systems

□ Communication via Event Subscription and Notification

- Support for heterogeneity
- Support for asynchronous communication



Middleware Approaches

■ Location Transparency

Figure 2.7 on Page 68



Method Invocation

■ Local vs Remote Invocation

Figure 5.12 on Page 223



Distributed Object Model

■ A Remote Object and Its Remote Interface

Figure 5.13 on Page 224

Distributed Object Model (Cont'd)

■ Instantiation of Remote Objects

Figure 5.14 on Page 225



RMI Components

■ Proxy

- Forwarding Messages to a Remote Object and Receiving the Reply
 - Making RMI transparent to clients

■ Dispatcher

- Receiving the Request and Selecting the Appropriate Skeleton Method

■ Skeleton

- Implementing Methods in the Remote Interface
 - Unmarshalling arguments and invoking the method

RMI Components (Cont'd)

■ Illustration

Figure 5.15 on Page 226

Remote Reference Module

Translating between Local and Remote Object References and Creating Remote Object References



RPC Components

■ Illustration

Figure 5.10 on Page 217